

Project Proposal

1 Group Information

Group Number: 6

Member List:

Logan Sichelstiel, Bryce Clement, Lucy Budde, and Robert Oliver

2 Introduction

Describe *briefly* what the project is about, answering the following questions:

- What is the title of your project?
- What is the goal of the project?
- What is the motivation for this project?
- Who are the customers/users?
- What development process will you be using?

Again, the introduction should be *brief but informative*. The introduction is where you write an overview for the readers to get an idea of what to expect in the proposal/report.

Our project is called Meal Coin. The goal is to create a marketplace where students can buy, sell, send, and receive meal swipes. The motivation behind Meal Coin is to bridge the disconnect between students who run out of meal swipes and those who have some left at the end of the week. The main customers will be students who frequently find themselves in situations where they have too many or are out of meal swipes at the end of the week. We will be using a Scrum development process with agile values, with four planned sprints along the way.

2 Novelty

Describe the novelty of the project. For example, what's new about the project compared to existing Software? Having a fancy UI is NOT a novelty. A novelty in research/project refers to the introduction of a new idea or a unique perspective that adds to the existing knowledge in a particular field of study. It involves bringing something fresh and original to the table that has not been done before or exploring an existing topic in a new and innovative way¹.

For this project, select two comparison software programs. Study the software and develop an idea of how you can do better than them. Again, simply having a fancy button is not a novelty. It needs to be important, such as explaining why implementing new features that are lacking in the existing software is important.

The novelty of our project is that we are combining a craigslist style peer to peer marketplace, along with blockchain-based authenticity for the campus meal swipe system. In the traditional Craigslist system, a peer-to-peer marketplace is allowed, but there is no verified ownership of the users' listings. Meal coin will require users to hold a meal swipe in their account in order to be able to make a listing. The smart contracts in the blockchain will prevent fake listings of meal swipes, which will ensure that every transaction is legit. This will create a marketplace that is not relying on trust and instead enforces the irreversible rules of the smart contracts.

Another novelty of our system comes from the ability to transfer and receive meal swipes from other students, which is not currently possible with Davidson's dining system. Without a way to transfer swipes in the traditional meal swipe system, unused swipes go to waste rather than being put in the hands of students who will actually be using them. By implementing a transferable system while still holding the expiration date on the swipes every week, we preserve the rules of the traditional system while maximizing the use of meal swipes.

The importance of these novelties lies in the fact that we reduce the waste of meal swipes that happen in the current system while still upholding the reliability of the system by backing our transactions with smart contracts.

3 Customer Needs

- Who is the primary customer outside the team?
- Who are the secondary stakeholders?
- What do the stakeholders want? Why?
- What is their desired overall experience?

The primary customer outside of the team will be the students at Davidson College. Our project enhances students' dining experience as it expands how they are able to use their meal swipes. It allows them to waste fewer meal swipes and get meal swipes for a lower price when they run out. The desired overall experience is that they are able to have more autonomy with their meal swipes, because they paid for them as part of their tuition.

The secondary stakeholder will be Davidson's dining admin. Although this project is a prototype of a new dining system, we still need to consider the dining admin's needs. If this were to be implemented, the Davidson admin would want the system to work stably, predictably, and without failures. They will require that the system be stable and that the redemption of meals is done without complication, to reduce the amount of difficulty on the staff and students when redeeming swipes. They will want it to be predictable so that the swipes reset correctly every week, so they don't have any financial losses due to this system. Lastly, they will want it to be without failures so that there is no way they can be sued or blamed for the mishap of students while using the system.

Their desired overall experience is for students to be able to buy and sell meal swipes smoothly, and for redemption at dining locations to be instant.

3.1 User Requirements

Write at least 5 SMART user stories based on the stakeholders' needs and wants:

As a <role> I want <action> so that <benefit>

1. As a student going away for the week during the semester, I want to be able to list 6 meal swipes in under a minute so that I can convert my unused meal swipes into cash rather than them going to waste.
2. As a student who ran out of swipes during the week, I want to purchase 1 meal swipe and have it ready to use in under one minute so that I can use it at a dining service without having to use my credit card.
3. As a student looking to buy a meal swipe, I want to be able to see all active listings for meal swipes in 5 seconds so that I can choose the lowest option for my purchase.
4. As a dining worker, I want to be able to redeem one swipe for a meal directly from the student's meal swipe wallet and have that transaction recorded on the blockchain instantly so that all transactions are verifiable and work with the weekly expiration rules.

5. As a dining worker, I want all swipes in students' wallets to be burned at the rollover period on Saturday at 11:59 pm so that the expiration date is enforced without any extra work than with the traditional swipe.

3.2 Acceptance Tests

Write **at least 5** acceptance tests for the user stories using the template:

Given <some context> **when** <triggering event> **then** <expected outcome>

1. Given a student has 2 meal swipes left in their account for the current week, when the student submits a sell order for 2 swipes each at \$7, then the system should display the listing on the marketplace and lock the 2 swipes in escrow within 5 seconds.
2. Given a student has no meal swipes left in their account for the current week, when the student tries to create a listing on the marketplace for a swipe, then the system will send a message saying "No meal swipes valid to list," and the student will not have a listing sent to the marketplace.
3. Given that a listing exists on the marketplace for 1 swipe at \$6, when a buyer agrees to the price and clicks purchase in the app, then the meal swipe will be transferred from the selling student's wallet to the purchasing student's wallet, the seller will receive \$6, and the listing will be taken down from the marketplace.
4. Given a student has 1 or more valid meal swipes in their wallet, when the student goes to purchase a meal at a dining location and sends the meal swipe from the app, then the dining worker will receive confirmation that the student has paid for the meal, and that swipe will be burned from the student's wallet.
5. Given that the clock reaches 12:00 a.m. on Sunday, when the weekly epoch happens, then all the swipes locked in escrow and in the students' wallets will be burned and will no longer exist.

4 Project Goals

4.1 Customer Problems and Benefits

What customer problem have you chosen to address?

- In implementation-free terms, what user benefit will the system provide?
- How will the benefit support the customer's desired overall experience?

We have chosen to address the customer problem of running out of meal swipes or having unused meal swipes at the end of the week. The system will allow users to have more access to meal swipes and have the ability to make some money out of their unused meal swipes. The users will also be able to send and receive swipes from their friends directly if they choose to do so. These benefits will allow the customers to have a better dining experience at Davidson College by allowing them to have more freedom and uses for their meal swipes.

4.2 Measure of Success

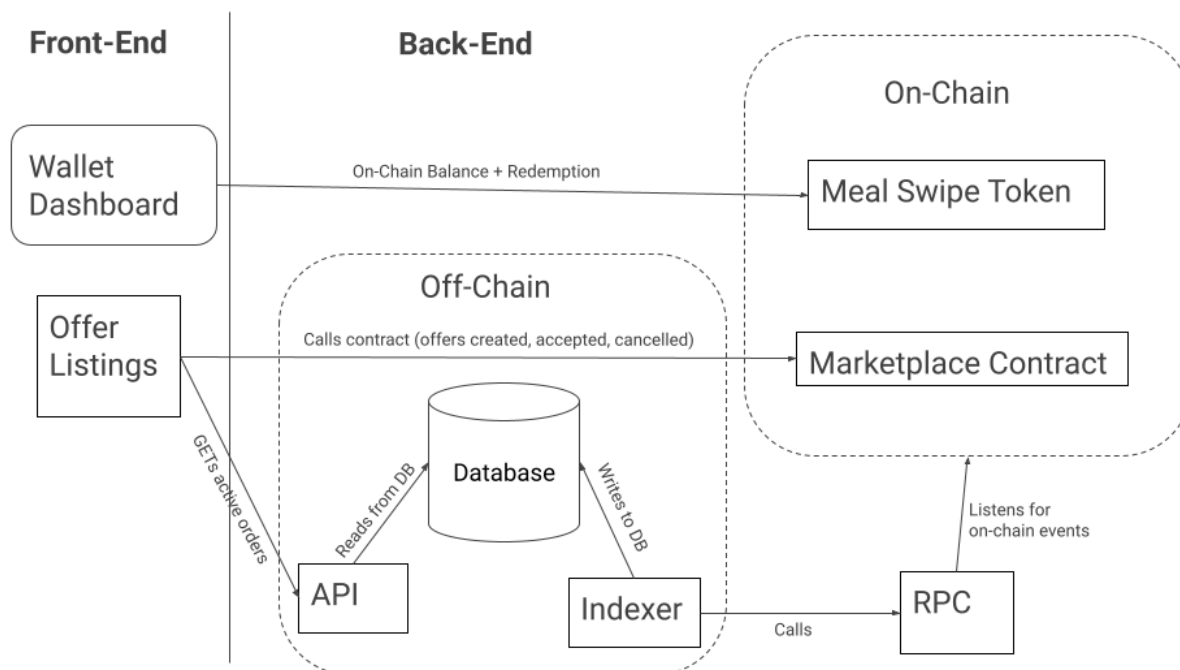
- Who outside the team have you tested the idea on?
- How will you know whether the customer got their desired benefits?
- What are your customer-centric measures of success?

We will test the idea on students outside of the team, who are on some sort of meal plan with Davidson, and have been in the situation where they have been out of or wasted swipes. We will know if the customer got their desired benefits based on the success of the transfer of the swipes and the speed of the user interface. We will also ask a few questions to the students after they test our system, asking how often they would use the system, and how happy they are with the overall process of buying, selling, and transferring swipes. Our customer-centric measures of success will be based on how easy and how fast students using this system have access to the marketplace and access to meal swipes.

5 System Description

For this proposal, a rough draft of this section is enough.

- Draw a block diagram to show how the proposed system will interact with external services, databases, etc. Clearly mark the system's boundaries.
- Use the above diagram to introduce the system.
- What are the main elements of the proposed system?



This is how our system will be designed. In the frontend, we will have 2 main pages: the wallet dashboard and the offer listings page. The wallet dashboard will display the user's balance of the meal swipe tokens. This will display the on-chain balance for the user's wallet address and the current week. The meal swipe token contract is a contract for the meal coin ERC-20 token. The contract will store the balances of the token for every address and week. It will also have a function for redeeming tokens when it is used at a dining, by burning one token from a user's wallet.

The offer listings page is where students can post buy and sell offers, and other students can accept. Every time a student posts an offer, it will call the marketplace contract that handles the marketplace logic. It will be an escrow contract that holds what the student is offering and allocates assets accordingly when an offer is accepted. For example, when a student posts a sell offering for a meal swipe at \$10, one of the

meal swipe tokens in their wallet will be sent to the escrow contract. Then, when another student accepts the offer, \$10 will be sent to the contract. The contract makes the necessary checks, then the \$10 goes to the seller's wallet, and the meal swipe goes to the buyer's wallet. The contract will also have a function to cancel orders and logic for offers expiring.

We will also have an indexer running in the backend that calls an RPC that listens for on-chain events like offers posted, accepted, and cancelled. Those events are written to our database. Then, when users check the offer listings page, our backend API will be called to get existing offers from the database. Offers will have expiry logic for the end of the week, so stale offers will not be displayed in the listings.

6 Solution Approach

For this proposal, a rough draft of this section is enough.

- Briefly describe how the system will work.
- What technologies (platform, tools, libraries, programming languages) will you use and why?
- How will you test and measure the adequacy of your test strategy?

Section 5 describes an overview of how the system will work. The contracts will be written in Solidity because it is the main standard used to write smart contracts. We will build and test the contracts with Foundry because it allows us to test functionality and edge cases quickly. We can test on a local chain simulation, and test the functionality of the escrows, week rollover behavior, offer cancelation and more. Once they are done, the contracts will be deployed to the blockchain. We plan to use Supabase to store things derived from on-chain events. Supabase uses Postgres, which is reliable and allows us to structure the database how we want. We decided to store the events in a database, so that we don't have to read the chain every time we want to see the offer listings. Our backend API will use Next.js and TypeScript. Since we plan to use the same for the frontend, that makes it easier to develop as it will be one codebase for the UI and API. Our API will be the data service layer that includes endpoints like GET /asks, GET /bids, GET /trades, and GET /redemptions, which read from our database. Our frontend will also use Next.js and Typescript. TypeScript is strongly typed, which will help us mitigate wrong type bugs and keep contract call shapes consistent. We will use an RPC provider like Alchemy, which allows us to read on-chain balances and events, which is crucial to the functionality of the project.

To test the functionality of our system, these are some of the tests we need:

- When offers are posted by the user on the listings page, we need to observe the contract calls and events emitted on the chain through a block explorer and ensure that the correct events are emitted.
- When events are emitted on-chain, we need to check if our indexer picks them up through the RPC by writing tests that output events picked up by the indexer.
- When the indexer picks up on chain events, we need to ensure that these are being correctly written to the database. We will write tests that simulate the indexer picking up test events and see if they are written to the database in the form we want them to.
- When opening the listings page, we need to compare it to the database to see if the frontend call to the API is correctly getting the updated offers. We can write tests that call the API and check if it is pulling from the database correctly.
- When a user cancels an order, we need to observe if the marketplace contract was correctly called

and check if the on-chain balances of the address were updated.

7 Project Management

- What development process will you use (Scrum, XP, Scrum+XP, etc.)?
- What are the reasons behind your choice?
- Describe your (brief) goals for each iteration (Proposal - Report 1, Report 1 - Report 2, and Report 2 - Final)

We will be using a Scrum development process. Since our project includes uncertain technical details, including smart contracts and escrow logic, Scrum fits our project the best, allowing us to build the project in small iterations, and test early to catch any bugs that we might not have seen until later on had we done a different approach. We will have a consistent schedule with 2-week sprints and weekly meetings on progress. We will have a main product backlog that includes all the key product requirements. Before every sprint, we will decide as a group on a sprint backlog to set product backlog items for the next sprint. After each sprint, we will have a sprint review where we get feedback from our stakeholders (students), review our progress, and plan for the next sprint.

These are our current goals for each iteration, and how they fit into the reports. They are subject to change as we assess progress after each sprint.

Sprint 1: We plan to have the smart contracts written and tested, so that the meal swipe token and marketplace escrow contract are set and ready for the later stages of development.

Sprint 2: We plan to have our frontend minimum viable product ready. This will include an easy flow to connect a wallet, balances shown in the wallet, and easy access to browse listings and also create listings. This is where we plan to be at Report 1.

Sprint 3: We plan to have the logic ready for the backend, holding the logic for the bids of meal swipe listings. This sprint will include having a working database/API for the listings and an indexer that reads on-chain events to write to the database. This is where we plan to be at Report 2

Sprint 4: This is our final sprint and will be the polished user interface and demo-ready application. There will be meal swipe redemption functionality, security checks, and all edge cases, including epoch rollovers, will be handled and accounted for. We plan to be at this stage for our final report.

8 Team Management

8.1 Roles

- What are the planned roles for the team members?
- What are the reasons for your decision?

Scrum Master: Bryce - Has the deepest understanding and knowledge of how the project should flow, will be the best person to keep us on track for our sprints and goals.

Product Owner: Logan - Knows how to interact with the customers well and get a good understanding of a product that the customers would like to use.

Developers: Everyone - We have a small team of only four, so we will need all hands on deck when it comes to developing this system.

8.2 Scheduling

- How often will the team meet?
- How will you meet as a team? Zoom? In-person?
- How often are you planning to meet your customer(s)?
- How will you meet with your customers? Zoom? In-person? Is this agreed with your customer(s)?

The team will meet once a week in person. If someone is away or cannot make a meeting, we will resort to making the meeting on Zoom. We are planning to meet with customers biweekly after each sprint in person to get the best understanding of what the customer wants, so we can iterate for the next sprint.

8.3 Background

Write brief introductions of each team member related to the experience and background of the project. For example, student A has experience developing Android mobile applications using Java. Simply having an experience in certain programming languages is fine as well. Consider this section an opportunity to determine who can be responsible for what part of the system (e.g., experience in JavaScript can do a front-end development while Java can do a back-end, etc.)

Bryce: Experience with TypeScript, Python, SQL, and Solidity. Has developed on-chain projects, built and deployed contracts, developed REST APIs, utilized RPC calls, and created web apps with Next.js and React. Also experience with databases.

Logan: Experience developing in Python, Java, JavaScript, and R. Has worked on projects doing frontend development and backend development, with more experience in backend development. Has connected backend with smart contracts in experience with projects.

Lucy: Experience with Python and Java, and a strong background in design and graphics. Has worked on projects doing frontend development and design.

Robert: Experience with Java, Python, C, and F#. worked on projects using Microsoft Azure and has heavy experience in backend development.

9 Constraints and Risks

- Are there any social, ethical, policy, or legal constraints?
- Will you have access to the data, services, and resources you need?
- Is there anything else you might need?

Constraints:

Our product includes the transferability of Davidson meal swipes. This is not currently allowed at Davidson, which may lead to some policy conflicts. Approval from Davidson Dining and administration would be required for this system to be implemented in real-world settings.

Ethical constraint: Since there will be a market where real money is involved, students could attempt to game the system somehow to profit off other students. However, the way our system is designed mitigates that risk because students who want to post their swipes at a fair price are able to, and students buying swipes can see those listings, so there will always be the chance to buy swipes at a fair price.

Due to the fact that this project is a prototype, we will not be required to have access to Davidson's

Dining Database. However, if we are able to get Davidson Dining on board with our project, we will have to have access to their Database, which could introduce some more ethical and legal constraints. All other resources for the project are available to the team, including all resources needed for blockchain and wallet components of the project.

Authentication: Since our product is only for Davidson Students we will need to implement a student verification system where students must sign in with their email, and the corresponding address will be mapped to them. The dining service team will have a master list of students and their corresponding addresses, so each student who has been verified will be loaded with 6 swipes at the beginning of the week. This ensures that there is only one wallet per student that swipes will be allocated to.

Campus policy risk: When looking at the Davidson College dining website, they mention that dining dollars are non-transferable. Although they don't explicitly mention anything about meal swipes, they may be opposed to it. However, our project is intended to be a prototype of what such a system would look like, so the campus policies will not block us from building the project.

Tax risk: Since there is money changing hands when meal swipes are bought and sold, there is a tax consideration that needs to be considered. Since meal swipes are valued at \$12 in our current campus system, we can have a cap at \$12 for the price the swipes can be sold for. This can mitigate the risk of incurring taxable income, because no one can make a profit from our system. There is a sales tax that may need to be implemented, which we can do by including a tax fee when selling meal swipes, which our system can automatically calculate and deduct from the money that goes to the seller.

Trust risk: Students may be worried about things like fake bids and offers. Our system mitigates that because students who post offers will have their swipes and dollars held in the escrow contract until the offer is accepted, cancelled, or expired. So fake bids are not possible, because those who are posting the offer must give up what they are offering when posting it.

10 Values

- What are the important principles (≥ 3) that will guide your team?
- For each principle, write a short description of what the principle means. For example,
Simplicity over complexity: We value simple and understandable solutions. When multiple approaches are possible, we prefer the one that is easier to maintain, test, and explain.
- Justify why each principle makes sense for your team and project.

Reduce Waste and Improve Access:

We want to create a system that minimizes unused meal swipes and maximizes access to these meal swipes for students who run out.

The primary goal of our project is to reduce the amount of wasted meal swipes at the end of each week and put them into the hands of the students who need them most at a fair price. This makes sense for our team because it is the motivation behind why we are creating this project.

Trust and Transparency:

We will prioritize a system that is verifiable, auditable, and fraud-resistant. Users will be able to fully trust

that the listings they see are real and backed by meal swipes.

We are dealing with finances and access to food, so making sure that this system is highly resistant to fraud is crucial. This value will keep our team driven to make this system as secure as possible, so the students can trust the listings they find.

Paced Improvement and Adaptability:

We value progress done in increments, along with true feedback from our customers, to adapt our project in order to fit the customers' needs.

Our system involves many different components, including on-chain and off-chain logic. So, having this value keeps the creation of this project on a structured and on track. This makes it easier to adapt and problem-solve when challenges arise.

11 GitHub

Follow the instructions below:

1. One person creates a GitHub repository.
 2. Add other members to the repository as collaborators.
 3. Update your README.md file with the title of your project, your group number, and the group members with their roles. Example below:
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1. Add me (username: hlim1) as a collaborator to the repository.
 2. Create a folder called “Reports” in the main branch.
 3. In the folder, add this proposal. You will add all the reports to this folder, in addition to submitting them to Moodle.
 4. Take a screenshot of the ‘Collaborators’ page and add it to this section.

